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Items assignment optimization for complex automated picking Systems

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Abstract

Order picking operation is the most expensive and time consuming process of all assignments in distribution center. The high efficient and low consumption automated systems, especially complex automated picking systems have been widely promoted and used. How to improve the efficiency and accuracy of the automated picking system becomes an important issue for improving the business capability of the distribution centers. By analyzing the operating mode of complex automated picking systems, the model of complex automated sorting systems is established and the operation time model is constructed based on the serial order picking strategy with the sequence from right to left. The items assignment optimization models for different types of picking machines are constructed, minimizing total picking time. The improved niche genetic algorithm is designed. In order to improve algorithm convergence speed, the k-means clustering method is used and the results are regarded as the initial population of clustering algorithm. By restricting the number of chromosomes whose location distribution of a certain item is fixed, the niche elimination operation process is improved and the diversity of population is maintained. Through the simulation analysis, the items assignment optimization algorithm leads to 7.5% decrease of the total order picking time, in comparison with the results determined by the traditional *EIQ-ABC* method. On the basis of items assignment optimization simulation experiments, items configuration and sort comparison tests in a multi-items picking area are designed, again validating the effectiveness.

Keywords Complex automated picking systems · Items assignment · Order picking strategy · Niched genetic algorithm

1 Introduction

The order picking operation has become the most expensive and time-consuming business in the distribution center [1–3]. If the operation efficiency of the distribution center is intended to be further enhanced, the main focus should be given to reduce the operation time of the order picking systems. Recently, the automated order picking systems suitable for handling multi-type items and smaller orders have gradually replaced the manual order picking systems [4,5]. The automated order picking systems greatly reduce the order picking time, labor intensity and improve the order picking accuracy. The automated order picking system, which is suitable for picking the items with basically consistent the dimensions, is generally composed of A-shaped sorting machines, horizontal sorting machines, conveyor systems and packaging systems.

How to improve the accuracy and efficiency of the automated order picking system has become an important issue, faced by distribution centers. Under the premise of the defined structural characteristics of the orders, the efficiency of automated order picking systems is mainly affected by its operating parameters, structure elements and order picking strategy. Recently, the operating efficiency of stand-alone order picking machines has been considerably improved. However, the operation efficiency of stand-alone picker is largely influenced by limited production capacity and expensive manufacturing cost. When the stand-alone efficiency reaches a specific level, it becomes difficult to be further improved. Based on this, improving the system efficiency by optimizing the layout elements, order picking strategy and

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parameters match of the automated order picking systems has become a new research hotspot [6–10].

Currently, the number of studies on complex automated picking systems (CAPS) have gradually increased. The CAPS generally consists of two or more sorting machines. Combining different order picking machines is used to achieve higher performance. The total order picking time is the key indicator to measure the performance of the automated order picking system. The total order picking time is calculated since the first picked item in the first order enters the picking system, until the moment that the last item in the last order leaves the delivery system [11,12].

On the basis of the defined order structure, the items assignment optimization improves the order picking efficiency without increasing operations cost. Regarding the items assignment issues in the double order picking zones, Zhang et al. [13,14] and Wu and Zhang [15] establish a mathematical model, starting the order picking ahead of time. But the mathematical model is only designed for the specific double picking zones in an automated order picking system, which are not applicable for the entire complex automated picking systems.Based on the serial order picking strategy with the sequence from right to left in complex automated order picking systems, the item picking position assignment model is constructed, and lastly solved through the improved niche genetic algorithm in this study.

2 Overview of complex automated picking systems

Since the characteristics of ordered items vary greatly, a complex automated picking system generally consists of single and multi-item picking machines.

The single item picking machine only picks one item within one picking action. It is featured by the simple action and small space; while its relatively low operation efficiency. For instance, the demand quantity for certain items in an order is k, then the single item picking machine requires k times to complete the demand quantity. Multi-item picking machines can pick out $1 - n$ items, or fixed m items in one action. The former is relatively widely used. Compared with single item picking machines, it covers a larger area and operates more complexly. A multi-item picking machine, in addition to the general picking actions, usually needs the automated replenishment operation. Its advantage is the relatively higher picking efficiency. For example, the demand quantity for certain items in one order is k $(1 < k < n)$, a multi-item picking machine can finish the order picking in one movement. In general, the single-item picking machine is suitable for items with smaller average demand quantity, while the multiple-item picking machine is suitable for items with larger average demand quantity.

There are multiple forms of complex automated picking systems. The basic type generally includes single item sorting units, multiple-item sorting units, delivery units, packaging units and control units. Fig. 1 shows the basic composition structure of a complex automated picking system.

The system generally takes the serial order picking strategy in the sequence from right to left. When fulfilling an order, single and multi-item picking machines will catapult items to the delivery system based on the order demand quantity; and then the delivery system will transport the items to the packaging unit, packing the items according to the order size.

3 Modeling of order picking operation time

3.1 Basic assumptions of system

In order to facilitate research, assumptions for the basic complex automated picking systems are listed as follows:

(1) Each picking machine is only responsible for one kind of item and there are no picking machines responsible for the identical item.
(2) The dimensions of the picked commodities should be basically identical.
(3) One movement of the single-item picking machine can only catapult one item, and the time to catapult one item is identical; the multi-item picking machine can catapult 1–5 items in one movement based on the demand quantity.
(4) During the order picking process, there is no out-of-stock.

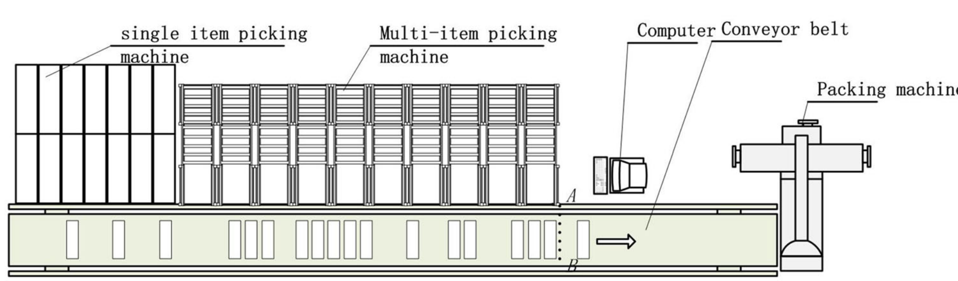

Fig. 1 Configuration of the complex automated picking systems

(5) Items must be placed together on the conveyor based on types and customer orders. There is no overlap and cross in the delivery system.
(6) The packing speed of the packing machine should match that of the picking machine and the conveyor.

3.2 Modeling of order picking operation time

The total number of picking items are set as n. As shown in Fig. 1, the picking facility is numbered from 1 to n in the order from right to left. The set of total items is denoted by N. A denotes the item set picked by the single-item picking machine; where the item type quantity contained is denoted as C_a. Set W denotes the item set picked by the multi-item picking machine, where the item type quantity contained is denoted as C_w; where, $A \subseteq N$, $W \subseteq N$, and $A \cap W = N$.

The total number of orders is set as O, a_{ij} is the picking quantity of item i in the order j. The multi-item picking machine is set to pick 1–5 items for one time. The time for a single picking movement is set as $\mathrm{t_w}$, and the item replenishment time is t_b.

The single-item picking machine is set to pick only one product for one time, and the single picking movement time is t_a. Because the replenishment operation does not affect the total order picking time, the time of replenishment in single-item picking machines is ignored.

In accordance with the serial order picking strategy, the picking time of a single order mainly consist of (1) the accumulated sorting operation time of sorting machines, and (2) the time of the remotest item delivered out of the picking zone.

For a given order j, the sorting operation time on its single-item picking machine can be expressed as:

$$T_j^a = \sum_{i=1}^{n} \eta_i \cdot t_a \cdot a_{ij} \tag{1}$$

where, the decision variable $\eta_i = \begin{cases} 1, & i \in A \\ 0, & i \in W \end{cases}$.

For multi-item picking machines, the number of picking movement of certain item in the given order j is decided by the margin of items on the picking operation surface of the picking machine. If the margin of items is greater than or equal to the demand quantity of the order j, then once picking movement can satisfy the picking requirements; if the margin is less than the demand, then the items on the operation surface should be replenished by the feeding movement of the items, and followed by the picking action again. The number of order picking item replenishment actions is decided by the demand quantity of the items. Thus, the sorting operation time on the multi-item picking machine for the order j can be expressed as:

$$T_{j\_w}^w = \sum_{i=1}^{n} (1-\eta_i) \cdot Sgn(a_{ij}) \cdot \left(ceil\left(\frac{a_{ij} - \mathrm{y}_{i(j-1)}}{5} \right) + 1 \right) \cdot t_w \tag{2}$$

where $\mathrm{y}_{i(j-1)}$ is the margin of item i on the picking operation surface of the picking machine after fulfilling order $j-1$; ceil (x) denotes the rounding in positive infinity direction of x direction; and $Sgn(a_{ij}) = \begin{cases} 1, & a_{ij} > 0 \\ 0, & a_{ij} = 0 \end{cases}$.

The replenishment time in multi-item picking machines will be included only when multiple picking actions occur in order to meet the picking quantity of a certain item in a certain order. The item replenishment time occurring in the gap of picking two orders will not be incorporated into the total picking time. Thus, the item replenishment time for the order j can be expressed as:

$$T_{j\_b}^w = \sum_{i=1}^{n} (1-\eta_i) \cdot Sgn(a_{ij}) \cdot \left(\left(ceil\left(\frac{a_{ij} - \mathrm{y}_{i(j-1)}}{5} \right) + 1 \right) - 1 \right) \cdot t_b \tag{3}$$

Based on (2) and (3), the total sorting operation time for order j in the multi-item picking area is:

$$T_j^w = \sum_{i=1}^{n} (1-\eta_i) \cdot Sgn(a_{ij}) \cdot \left(\left(ceil\left(\frac{a_{ij} - \mathrm{y}_{i(j-1)}}{5} \right) + 1 \right) \cdot (t_w + t_b) - t_b \right) \tag{4}$$

The time for delivering the remotest item from the picking zone depends on (1) the conveying speed of the conveying system, (2) the width of single-item picking machine and multi-item picking machine, and (3) the position of the remotest item. The zone of the remotest items in the order j can be calculated through the following parameters:

$$\theta = \max_{1 \le i \le n} \left(\mathrm{Sgn}(a_{ij}) \cdot d_i \right) - \sum_{i=1}^{n} (1-\eta_i) \tag{5}$$

where d_i is the location of item i in the picking systems, and the value is equal to the number of the picking machine. If $\theta > 0$, then the remotest item is located in the single-item picking machine zone. Vice versa, the multi-item picking machine zone. The sorting equipment is set to be aligned uniformly, in which the width of the single and multi-item item picking machine occupy along the direction of transmission of l_a and l_w. The time of the remotest item to convey out of the picking zone can be expressed as:

$$T_j^s = \frac{Sgn(\theta) \cdot \max(-\theta \cdot l_w, \theta \cdot l_a) + \sum_{i=1}^{n} (1 - \eta_i) \cdot l_w}{v} \tag{6}$$

Based on (1), (4), and (6), the picking time of order j can be obtained as:

$$T_j = T_j^a + T_j^w + T_j^s \tag{7}$$

The total picking time in one batch is expressed as:

$$T = \sum_{j=1}^{O} (T_j^a + T_j^w + T_j^s) \tag{8}$$

Operating parameters (t, v, l) are denotes as X. The function relationship of the total picking time in complex automated picking systems can be solved:

$$f(X) = \sum_{j=1}^{O} (T_j^a + T_j^w + T_j^s) \tag{9}$$

3.3 Model analysis

As known from formulas (1), (4), (6), and (8), the total order sorting operation time of picking system is mainly related with the picking quantity, and the delivery time of the remotest items.

Proposition 1 *When complex automated picking systems adopt the serial order picking strategy from right to left, its operating time $T_j^a + T_j^w$ is only related to the assignment policy, and is unrelated to its sorting sequence in the item picking zone.*

Proof For items Set A picked in single-item picking machines, formula (1) can be transferred to

$$T_j^a = \sum_{i \in A} a_{ij} \cdot t_a \tag{10}$$

where $\sum_{i \in A} a_{ij}$ is related to the composition of the Set A. When the items in the Set A are defined, it is a constant value.

For items Set W picked in multi-item picking machines, formula (4) can be expressed as

$$T_j^w = \sum_{i \in W} Sgn(a_{ij}) \cdot \left(\left(ceil\left(\frac{a_{ij} - y_{i(j-1)}}{5} \right) + 1 \right) \cdot (t_w + t_b) - t_b \right) \tag{11}$$

where the items $i \in W$, and $y_{i(j-1)}$ can be expressed as

$$y_{i(j-1)} = \begin{cases} 5 & , j = 1 \\ 5\text{-mod}\left(\frac{\left(\sum_{j=1}^{j-1} a_{ij} \right) - 5}{5} \right) & , j > 1 \end{cases} \tag{12}$$

where mod (x, y) denotes as modulo operation. □

It can be known from the formula (12) that the value of $y_{i(j-1)}$ is only related to the items a_{ij} and is unrelated to the sorting sequence. Therefore, it can be known that the value of formula (11) is only related to the item composition of Set W.

To sum up, the picking operation time $T_j^a + T_j^w$ is only related to the item assignment condition in the Set A and W, and is to the sorting sequence in the picking zone.

Corollary 1 *When a complex automated picking system adopts the picking strategy from right to left and items zone is defined, the picking operation time $T_j^a + T_j^w$ is only related to the number of items picked, and is unrelated to its sorting sequence in the picking zone.*

Proof based on the formulas (10), (11), and (12), for the defined zones A and W, the picking operation time $T_j^a + T_j^w$ is only related to the items picking quantity, and is unrelated to the items sorting sequence in the picking zones.

Proposition 2 *When a complex automated picking system adopts the picking strategy from right to left, the time of the remotest item conveyed out of the picking area T_j^s is related to the items assignment policy and its location in the picking zones, and is unrelated to the picking quantity.*

Proof Assume that the total items quantity is set as w in the Set W, namely $\sum_{i=1}^{n} (1 - \eta_i) = w$, the location of the remotest nonzero items in the order j is denoted as $d_j^{\max}$, namely $\max_{1 \le i \le n} \left(\text{Sgn}(a_{ij}) \cdot d_i \right) = d_j^{\max}$. By combining formulas (5), (6), T_j^s can be expressed as:

$$T_j^s = \begin{cases} \frac{d_j^{\max} \cdot l_w}{v}, & w \ge d_j^{\max} \\ \frac{\left(d_j^{\max} - w \right) \cdot l_a + w \cdot l_w}{v}, & d_j^{\max} > w \end{cases} \tag{13}$$

(1) T_j^s is related to the items assignment policy in the picking zone

Assume the sorting sequence is made as the same, namely $d_{j,w=N}^{\max} = d_{j,w=0}^{\max}$. When $w = N$, that is, the picking systems are completely composed by multi-item picking machines: $T_{j,w=N}^s = \frac{d_{j,w=N}^{\max} \cdot l_w}{v}$. When $w = 0$, namely the picking systems are completely composed of the single-item picking machines: $T^s_{j,w=0} = \frac{d^{\max}_{j,w=0} \cdot l_a}{v}$. From the condition $l_w > l_a$, it can be obtained as $T^s_{j,w=N} > T^s_{j,w=0}$. Thus it can be known that T^s_j is related to the items assignment policy in the picking zone.

(2) T^s_j is related to the sorting sequence in the picking zones

Assume the items assigned in the W and A are made to be constant, and the quantity of nonzero items in the single-item picking zone and multi-item picking zone of order j are $C^j_{a\_nz}$ and $C^j_{w\_nz}, C^j_{a\_nz} \leq C_a$ and $C^j_{w\_nz} \leq C_w$. When the remotest items in order j are located in the single item picking zone, its possible number of position is $\left(C_a - C^j_{a\_nz}\right) + 1$. When the remotest items in order j are in the multi-item picking zone, its possible number of position is $\left(C_w - C^j_{w\_nz}\right) + 1$. To sum up, in whatever cases, the possible positions of the remotest items of order j are all greater than or equal to 1; namely, the value $d^{\max}_j$ will change with the sorting sequence of the items in the formula (13). Therefore, T^s_j is related to the items sorting sequence in the picking zone.

(3) T^s_j is unrelated to the amount of items picked

Based on formulas (5) and (6), during the solution process of T^s_j, whether the certain item is the nonzero or not is defined solely based on the sign function. Thus, T^s_j is unrelated to the picking quantity. □

Corollary 2 *When a complex automated picking system adopts the picking strategy from right to left, for the defined sorting sequence in an item picking zone,* $\sum_{j\in O} T^s_j$ *shows the positive correlation with the ordered times of the remotest items, and unrelated to the picking quantity.*

Proof when the items assignment policy and sorting sequence is defined, for the given O orders, it is assumed that there are M items used as the remotest items, which can be expressed as f_1, $f_2 \ldots f_M$ according to the sequence from right to left. The delivery time of the remotest items can be expressed as:

$$\sum_{j\in O} T^s_j = k_1 \cdot T^s_j\Big|_{f_1} + k_2 \cdot T^s_j\Big|_{f_2} + \cdots + k_M \cdot T^s_j\Big|_{f_M} \quad (14)$$

In this formula, k_t represents the times of item f_t treated as the remotest items in O orders, where $1 < k_s < N - M + 1$ and $\sum_{t=1}^{M} k_t = O \cdot T^s_j\Big|_{fs}$ denotes the delivery time when f_t is the remotest item. □

As known from assumptions and formula (13), $T^s_j\Big|_{f_{t+1}} > T^s_j\Big|_{f_t}$. Based on formula (14), $\sum_{j\in O} T^s_j$ increases with ordered times as the remotest item, as known as k_M. From the proof of Proposition 2, T^s_j is unrelated to the number of items picked.

4 Items configuration optimization model

Based on the order picking operation time model, three issues should be solved to improve the optimal efficiency of complex automated picking systems: (1) how to determine the quantity of single- and multi-item picking machines respectively; (2) how to assign the items in two picking zones, and (3) how to arrange items in the picking zones. Among which, the first two issues can be combined as the items configuration problem, which belongs to the $0-1$ assignment planning problem. The total number of possible solutions is $\sum_{k=1}^{n} \mathrm{C}^{\mathrm{k}}_{\mathrm{n}}$, which is an NP-hard problem. The third question is the item sorting problem, which also belong to one specific category of assignment problems. The number of possible solutions is $P^w_n \cdot P^a_n$ in the case of defined items zoning policy, which is also an NP-hard problem.

As known from Proposition 1 and Corollary 1, the system picking operation time $T^a_j + T^w_j$ is related to the optimization of items configuration, and is unrelated to the items sorting problem in the picking zone. Hence, the main objective for the items configuration optimization is to reduce the system picking operation time. As known from Corollary 2, for the system with the defined items configuration, the transmission time $\sum_{j\in O} T^s_j$ of the remotest items is positively correlated with the ordered times. Therefore, the sorting sequence of ordered times along the transmission direction in ascending order is a better solution for the problem. In order to simplify the problem, based on the item configuration optimization, this paper makes the arrangement of the items assigned in the picking zone in the ascending order from left to right based on the ordered times, and compare the final optimized results with the traditional arrangement solution in the ascending order based on the picking quantity. If it is better than the latter, then no further optimization will be made. From the above analysis, the objective function of this problem can be defined as:

$$\mathrm{F(X)} = \min \mathrm{f(X)} = \min \left(\sum_{j=1}^{o} T^a_j + T^w_j + T^s_j\right) \quad (15)$$

where $\eta_i \in \{0, 1\}$.

5 Model solution

The target problem is an NP-hard problem and can only be solved by heuristic algorithms. There are a large number of feasible solutions for the optimization model. The solution algorithm should not only have the excellent local search capability, but also the sound level of global search. In this paper, the model solution is conducted through the improved niche genetic algorithm based on k-means clustering[16].

(1) Chromosome code design

Arrange the value of n pieces of η_i in order, which is the binary coding with equal length. The coding string corresponds to the original sorting sequence of items $1-n$ from right to left, wherein 1 denotes the corresponding item in the single-item picking zone, and 0 denotes the corresponding items in the multi-item picking zone.

(2) The initial population structure based on k-means clustering

He et al. [17], and Tang and Liu [18] suggest that the selection of the initial population directly impacts the global convergence and search efficiency of genetic algorithms. Therefore, in order to form the initial population, the algorithm is added in part with the individuals containing specific chromosome fragments based on the random uniform population generation.

As known from Corollaries 1 and 2, the picking quantity of items and the frequency of order are important influence factors. The items with smaller number of picking items and ordering frequency are appropriate to be placed in the single-item picking zone, or in multi-item picking zone vice versa. The k-means clustering method, which is proposed in Han et al. [19] to generate part of the individuals in the initial population, can be used according to the following steps:

Step 1 Solve the picking quantity and frequency of ordered of various items in the order, which are denoted as vectors V_Q and V_T respectively. The picking amount should be maximum-minimum standardized subject to the following linear data transformation formula:

$$VQC = \frac{V\_Q - \min V\_Q}{\max V\_Q - \min V\_Q} \cdot (\max V\_T - \min V\_T) + \min V\_T \tag{16}$$

Step 2 Let matrix $X = [V\_T; VQC]$, and then the k-means method is used to divide X into $k(3 < k < n)$ clusters, and solve the center point $c_1, c_2 \cdots c_k$ of k clusters.

Step 3 Calculate the distance between the center points of k clusters and the point $(0, 0)$. Take the items contained in the center point of the minimum of $\mathrm{fix}(\frac{k}{4})$ (rounded towards zero direction), and set the position of corresponding items in clusters to 1, while the remaining items position to 0, so as to constitute an individual.

Step 4 Change k value, and repeat steps 2 and 3 until obtaining the desired number of individuals.

Step 5 Put the generated individuals from step 4 into the initial population that is randomly generated, and replace the individuals of the corresponding number randomly.

(3) Fitness function

The optimization model is to minimize the objective function. The fitness function should be set as follows:

$$G(\mathrm{X}) = \begin{cases} C_{\max} - f(\mathrm{X}), & f(\mathrm{X}) < C_{\max} \\ 0 & f(\mathrm{X}) \geq C_{\max} \end{cases} \tag{17}$$

s.t.

$$X = (t, v, l) \tag{18}$$

$$C_{\max} = \max(f(\mathrm{X})) \tag{19}$$

(4) Genetic manipulation

The proportional selection method based on the optimal preservation strategy is selected. It first proportionally produces the next generation of the population, then randomly replace the individuals of the worst fitness in the generated population with the best fitness-value individuals among the previous population.

The uniform crossover method is used to exchange the corresponding gene from the pairwise individuals with the same crossover probability Pc so as to form the new individuals.

The basic bit mutation method is used to change the value of the certain gene locus with mutation probability P_m.

(5) Niche-out operations

The new population after the genetic manipulation operation and K individuals retained from the last population is merged into a new group $P(M+K)$ with a number of $M+K$ individuals.

The conventional niche-out operation [20] compares the *Hamming* distance between individuals in the new groups above. If the distance is less than the predetermined distance L, then compare their fitness values, and apply a penalty function to the individual with the lower fitness value, so as to greatly reduce its fitness. This research appropriately improves the niche-out operation. First, decode the individuals in the large population *P (M + K)*. Secondly, compare pairwise the corresponding coding positions of the individuals. Lastly, record r, namely the number of gene locus n which values are not the same in the same coding bit. Set a fixed value R, $R < M$. If $r < R$, then the penalty function should be applied to the individuals with smaller fitness in order to continually reduce its fitness. This method can help limit the number of chromosomes when the assigned items are fixed at certain positions, and further retain the best individuals so as to maintain the diversity of population.

The generated $M + K$ individuals with new fitness values are arranged in the descending order. Select the first M individuals to form the new population.

(6) Termination condition

As there are many feasible, a larger number of iterations D is used as the termination condition. If termination condition is not met, then repeat steps (3)–(6).

6 Simulation experiment

6.1 Design of simulation experiment

In order to validate the effectiveness of proposed items optimization method and the solution algorithm in complex picking systems, a complete order data in a working day is chosen from the aforementioned tobacco distribution center for the simulation analysis. There are 825 orders, including 103 kinds of items, and 42,711 items. The parameters of automated picking system is denoted in Table 1:

Table 1 The parameters of automated picking system

Parameter	Value	Parameter	Value
The speed of the conveyor transport system	1 m/s	The distance between two single-item picking machines	0.2 m
The single sorting operation time	0.3 s	The distance between two multi-item picking machines 0.5	0.5 m
The single replenishment time	0.3		

Table 2 The parameters in the niche genetic algorithm

Parameter	Value	Parameter	Value
Pc	0.9	Niche exclusion factor CF	3
Pm	0.001	The number of excluded individuals K	10
Population quantity M	150	The distance parameter between niches L	20
The number of iterations D	20		

Matlab7.4 is used to solve this optimization problem. The parameters in the niche genetic algorithm are set as follows in Table 2. In order to ensure the accurateness of test results, simulation experiments for optimization are conducted in ten times.

In order to verify the effectiveness of the items assignment optimization method for the complex automated picking systems, the optimization method designed in this paper (#5) is analyzed, in comparison with the following four methods respectively:

(1) Traditional items assignment methods (# 1): Items of category *A* are assigned to multi-item picking zone in reverse order, and items of categories *B* and *C* are reversely assigned to single-item picking zone.
(2) The traditional items assignment methods (# 2): Items of categories*A* and *B* are assigned to multi-item picking zone in reverse order, and items of category *C* is reversely assigned to single-item picking zone.
(3) Method under extreme conditions (# 3): all items are assigned to the single-item picking zone in descending order according to the order quantity.
(4) Method under extreme conditions (# 4): all items are assigned to the multi-item picking zone in descending order according to the order quantity.

The simulation results are shown in Table 3.

6.2 Results analysis

6.2.1 Intuitive analysis

According to the simulation results, the relations between the number of assigned items in multi-item picking zone and the total picking time are plotted in Fig. 2. The total order picking time of the items assignment optimization method

Table 3 Simulation results of the total picking time of CAPS under the different items assignment methods

No.	Items assigned in multi-item picking machines	Items assigned in single-item picking machines	Total order picking time (second)
# 1	1, 68, 6, 43, 49, 96, 57, 87, 21, 15, 66, 46, 3, 71, 19, 38, 97, 5, 86, 45, 40	The remainder are sorted in reverse order based on *ABC* classification	16865.2
# 2	20, 32, 60, 39, 52, 30, 54, 25, 77, 91, 27, 90, 82, 98, 47, 42, 59, 92, 73, 94, 56, 44, 101, 7, 8, 100, 13, 58, 17, 16, 34, 1, 68, 6, 43, 49, 96, 57, 87, 21, 15, 66, 46, 3, 71, 19, 38, 97, 5, 86, 45, 40	The remainder are sorted in reverse order by *ABC* classification	18850.3
# 3	Null	Arranged in the ascending order based on order quantity	17674.1
# 4	Sorted in the ascending order based on order quantity	Null	18550.7
# 5	86, 19, 38, 3, 5, 45, 71, 46, 97, 40	The remainder are arranged in the ascending order based on order quantity	15595.8

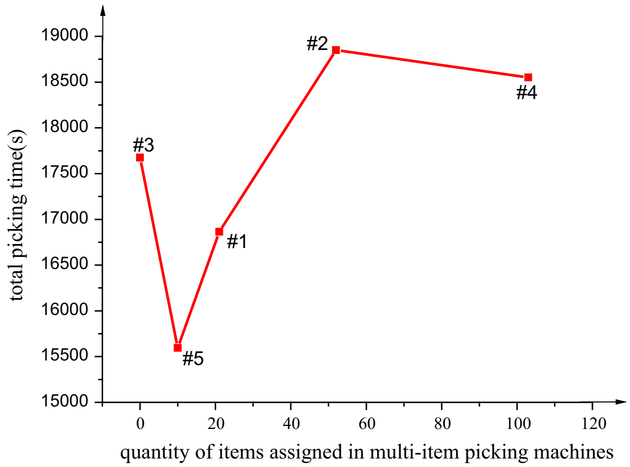

Fig. 2 Comparative analysis of simulation results

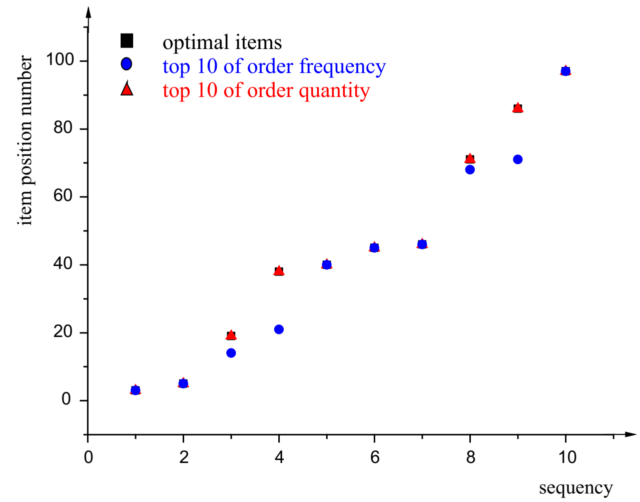

Fig. 3 Comparison of items in multi-item picking zone after optimization and special items of the same quantity

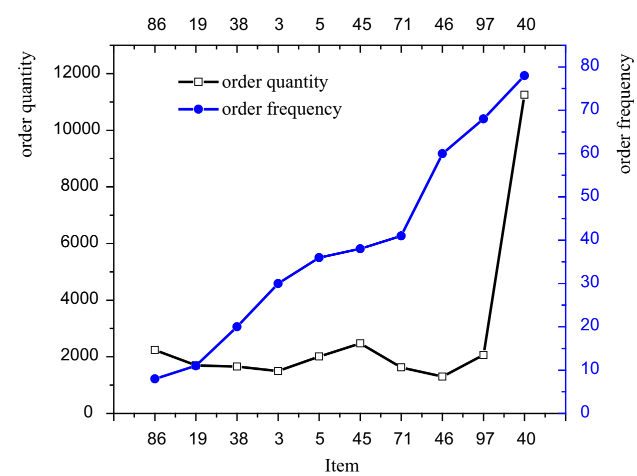

Fig. 4 Characteristic analysis of items in multi-item picking zone after optimization

designed in this paper is the lowest, which is 7.5% lower than the traditional program assigning the items of category *A* to the multi-item picking zone, and is 17.3% lower than the program which assigns items of categories *A* and *B* to the multi-item picking zone.

In the comparative experiments, two programs which assign all items to single-item picking zone or multi-item picking zone respectively are the special cases, of which the total picking time are both less than that in the program which assign the items of categories *A* and *B* to the multi-item picking zone, and both are greater than the items assignment optimization program and the one assigning items of category *A* to the multi-item picking zone. It results from the characteristics of two picking machines, which is in line with the initial analysis results. Meanwhile, the traditional program (#1) which assigns the items of Category *A* to the multi-item picking zone are all smaller than two exceptional cases, indicating that the program also belongs to one local optimal assignment program.

6.2.2 Characteristics analysis of items in multi-item picking zone after optimization

Table 3 lists the indexes of ten items assigned in the multi-item picking zone and the sorting sequence after the niche genetic algorithm. The ten items obtained after optimization are compared with the top ten items sorted in the descending order based on the order quantity and order frequency, which are as shown in Fig. 3 (sorted in the ascending order of their items number). These results coincide with the top ten items of the most ordered quantity, and six out of ten items with the most ordered frequency.

As known from and the propositions, the sorting sequence of ten items obtained from the proposed optimization program is set according to the ascending order of the order frequency, which is shown in Fig. 4.

To sum up, the items obtained from proposed optimization program are the top ten items sorted in the ascending order of order quantity. These items are sorted in the picking zone in the ascending order of the order frequency.

6.3 Verification of optimization results

During the solving process, in order to facilitate operation, only the items assignment problem has been optimized. The sorting sequence in the sorting zones is set in the ascending order of the order frequency. From the analysis above, the ten items obtained from the optimization program are the top ten items of the largest order quantity. In order to verify the effectiveness, the following five simulation experiments are designed to analyze the influence of order quantity, order frequency and the *ABC* classification.

(1) According to the descending order of *ABC* classification, 1–102 items are assigned to the multi-item picking zone. The remainder are assigned to the single-item picking zone. The items in each picking zone are sorted in the ascending order based on *ABC* classification.
(2) According to the descending order of the order quantity, 1–102 items are assigned to the multi-item picking zone. The rest are assigned to the single-item picking zone. The items in each picking zone are sorted in the ascending order based on the order quantity.
(3) According to the descending order of the order frequency, 1–102 items are assigned to the multi-item picking zone. The rest are assigned to the single-item picking zone. Products in each picking zone are sorted in the ascending order based on order frequency.
(4) According to the descending order of the order quantity, 1–102 items are assigned to the multi-item picking zone. The remainder are assigned to the single-item picking zone. The items in each picking zone are sorted in the ascending order based on the order frequency. Among them, the optimal solution based on the optimization model are included in this experiment.
(5) According to the descending order of the order frequency, 1–102 items are assigned to the multi-item picking zone. The rest of items are assigned to the single-item picking zone. The items in each picking zone are sorted in the ascending order based on the order quantity.

The optimal results of the above five groups are shown in Table 4, among which the optimal result (i.e. the 4th group) is that obtained through the optimization method in this paper. That is, in the multi-item picking zone, ten optimal items are selected in the descending order of order quantity, and sorted as order frequency. However, when these ten items are sorted in the ascending order of the order quantity (i.e., the second group), the total picking time is 16044.4 s, which is 2.9% more than optimal results. The number of the optimal items in the multi-item picking zone selected successively in the descending order based on *ABC* classification is nine and the total picking time is 15851.8 s, which is sub-optimal among five experiments, and nearly 2% more total picking time compared the optimal results. Compared with the traditional assignment method that the items of Category *A* are all assigned to the multi-item picking zone, the total picking time is reduced by 6%, and the number of items is reduced to 11. The number of optimal items in the second experiment is seven. Its total picking time is 15897.7 s, ranking the third. The number of items in the rest two experiments is both one. There are a wide gap between these results and the first three experiments in total picking time, but these two cases are still relatively dominated by the experimental result of sorted as order frequency (i.e. the 3rd group).

The results of five simulation experiments are shown in Fig. 5. The two curves that items are assigned according to the order frequency are similar, with larger oscillation in the front section and the stability in the rear section. The overall trend is that the total picking time increases with the number of items in the multi-item picking zone. At the same time, when the items are selected in the same descending order of order frequency, the total picking time by using the ascending order of order frequency is significantly superior to that in the solutions by using the ascending order of order quantity. The trends of the other three curves are similar., the total picking time is decrease first and then increases rapidly with the increase of items in the multi-item picking zone.

When the number of items assigned in the multi-item picking zone is more than 40, the increase rate of the total picking time becomes smaller. All three curves reach the optimum

Table 4 Selection of different items in the multi-item picking zone and the optimal results of sorting principle

No.	Select	Sort	Optimal items		
			Number	Sorting	Total picking time
1	Select items in descending order after *ABC* classification	Sort in ascending order after *ABC* classification	9	3, 71, 19, 38, 97, 5, 86, 45, 40	15851.8
2	Select items in descending order of order quantity	Arrange in the ascending order of quantity ordered	7	19, 38, 97, 5, 86, 45, 40	15897.7
3	Select items in descending order of ordered times	Sort in ascending order as ordered frequency	1	40	16036.4
4	Select items in descending order of order quantity	Sort as ordered frequency	10	86, 19, 38, 3, 5, 45, 71, 46, 97, 40	15595.8
5	Select items in descending order of ordered times	Sort as order quantity	1	40	16386.8

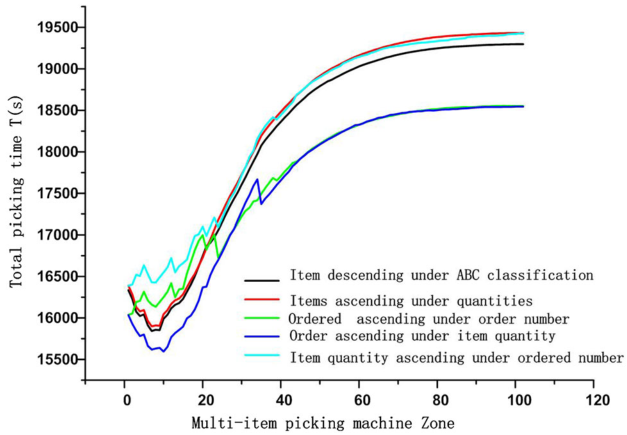

Fig. 5 Relation between the number of items of multiple items under the different sorting criteria and the total picking time

value in the interval of 6–10, and the optimal value is better than the optimal value of the other two experiments. These three curves stem from the solutions that items are sorted based on order quantity.

To sum up, for the items assignment problem in complex automated picking systems, the empirical method is that around top ten items in the descending order of the order quantity are assigned to the multi-item picking area, and all items in the picking zone are sorted in the ascending order of the order frequency.

7 Conclusion

(1) According to the model of basic automated picking system, the fundamental type of complex automated picking system is constructed. The operation time model of the basic complex automated picking systems is constructed based on the order picking strategy of the order from right to left. Through the analysis of operation time model, the items configuration optimization method of the basic complex automated picking system items is constructed. The niche genetic algorithm based on k-means clustering is designed to solve optimization model.

(2) Through the simulation analysis, the effectiveness of the proposed items configuration optimization method is proved. The obtained optimization results is remarkably better than the traditional items assignment program based on the *EIQ-ABC* classification.

(3) Through the results of analysis and optimization, the comparative experiments are designed based on the items configuration, which again prove the effectiveness of the optimization method. Meanwhile, the rule of thumb for items assignment optimization in the complex automated sorting systems under the set conditions is summed up. The top 10% items sorted by order quantity should be assigned to the multi-item picking area, and the rest of items are assigned to the single item picking area. The items in the picking area are sorted in ascending order of the order frequency.

(4) The research on picking strategy only considers the order picking of batch orders. The next step should consider the impact of order batches on the total picking time of the system based on order analysis.The discussion of system structure elements is not comprehensive enough and the next research should be a comprehensive and systematic study of the impact of different system components on the total picking time of the system.

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Compliance with ethical standards

Conflict of interest The authors declare that there is no conflict of interest regarding the publication of this paper.

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